

Topic 1.3 - Rates of Change in Linear and Quadratic Functions

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Use first and second differences correctly, including the equal-spacing requirement.
- Construct linear or quadratic models from values, rates, and structural conditions.
- Derive and interpret how average rates of a quadratic change across equal intervals.

Section A: Foundation and representation

Question 1

Foundation

No calculator

For inputs 0, 1, 2, 3, 4, outputs are 8, 13, 18, 23, 28. Classify and write a rule.

Question 2

Finite differences

No calculator

For outputs $-1, 2, 11, 26, 47$ at inputs 0, 1, 2, 3, 4, find a quadratic rule.

Section A continued

Question 3

Missing value

No calculator

A quadratic table for $x = 0, 1, 2, 3$ has outputs $4, 9, 18, k$. Find k .

Question 4

Nonuniform data

No calculator

Inputs $0, 2, 5, 9$ have outputs $1, 7, 28, 82$. Explain why raw differences do not establish degree and compute rates.

Question 5

Construction

No calculator

Determine the quadratic through $(0, -2), (1, 1), (2, 10)$.

Section B: Application and reasoning**Question 6**

Rate structure

No calculator

For $q(x) = 4x^2 - 7x + 2$, find rates on $[1, 3]$ and $[3, 5]$.

Question 7

Inverse reasoning

No calculator

A quadratic has consecutive one-unit rates $-5, -1, 3, 7$. Find its leading coefficient.

Question 8

Parameter

No calculator

For $f(x) = ax^2 + 2x + 1$, rates on $[0, 2]$ and $[2, 4]$ differ by 24. Find a .

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A student labels a table quadratic because output differences are 4, 8, 12. State the missing condition and give an input list that invalidates the claim.

Question 10

Challenge

No calculator

Construct a quadratic whose one-unit rate on $[x, x + 1]$ is $6x - 1$ and whose value at 0 is 5.

Question 11

Context

No calculator

A ball has height $H(t) = -5t^2 + 20t + 1$. Find one-second rates on $[0, 1]$, $[1, 2]$, $[2, 3]$ and interpret.

Question 12

Proof challenge

No calculator

Prove that a quadratic expression with zero second difference for nonzero step is actually linear or constant.

Section D: AP-style multiple choice

MCQ 1

No calculator

First differences are $-2, 3, 8, 13$ for unit steps. What is the quadratic leading coefficient?

- A. 2.5
- B. 5
- C. 10
- D. Cannot determine

MCQ 2

No calculator

Which table can be linear?

- A. Inputs 0, 1, 2; outputs 1, 4, 9
- B. Inputs 0, 2, 4; outputs 3, 9, 15
- C. Inputs 0, 1, 3; outputs 2, 6, 10
- D. Inputs 1, 2, 3; outputs 5, 5, 7

MCQ 3

No calculator

For $q(x) = -2x^2 + 7x + 1$, how do consecutive one-unit rates change?

- A. -4
- B. -2
- C. 2
- D. 4

MCQ 4

No calculator

Rates on $[0, 2]$ and $[2, 4]$ are 6 and 14. Find the leading coefficient.

- A. 1
- B. 2
- C. 4
- D. 8

AP-style free response 1

Calculator permitted - 6 points

Production values are

n	0	1	2	3	4
$P(n)$	12	18	28	42	60

A. Use differences to identify a model type.

B. Determine $P(n) = an^2 + bn + c$.

C. Find and interpret the rate on $[1, 4]$.

D. Predict $P(6)$ and state one caution.

AP-style free response 2**No calculator - 6 points**

A quadratic $q = ax^2 + bx + c$ has rate 3 on $[0, 2]$, rate 11 on $[2, 4]$, and $q(1) = 5$.

A. Write two equations from the rates.

B. Find a, b .

C. Find c and write q .

D. Find the second difference for step 3.
